PROJECT 5

GYM MEMBER

Simple Linear Regression with Single Predictor

PRESENTED BY MUNAWAR

BACKGROUND

- This analysis uses the Gym Members Exercise Tracking dataset, which consists of 973 gym members.
- The goal is to study the relationship between exercise duration (Session Duration in hours) and calories burned.
- The method used is Ordinary Least Squares (OLS)—a simple linear regression with one predictor.

EXPLORATORY DATA ANALYSIS

973	1.26 hrs	905.4	0
Data Sample	Avg. Session Duration (Hour)	Avg. Calories Burned (kal)	Missing Value

Variables

Predictor (X)

- Session Duration (hours)

Outcome (Y)

- Calories Burned

Goal

- Study the relationship between exercise duration and calories burned

OLS LINEAR REGRESSION MODEL

Simple regression built in this equation

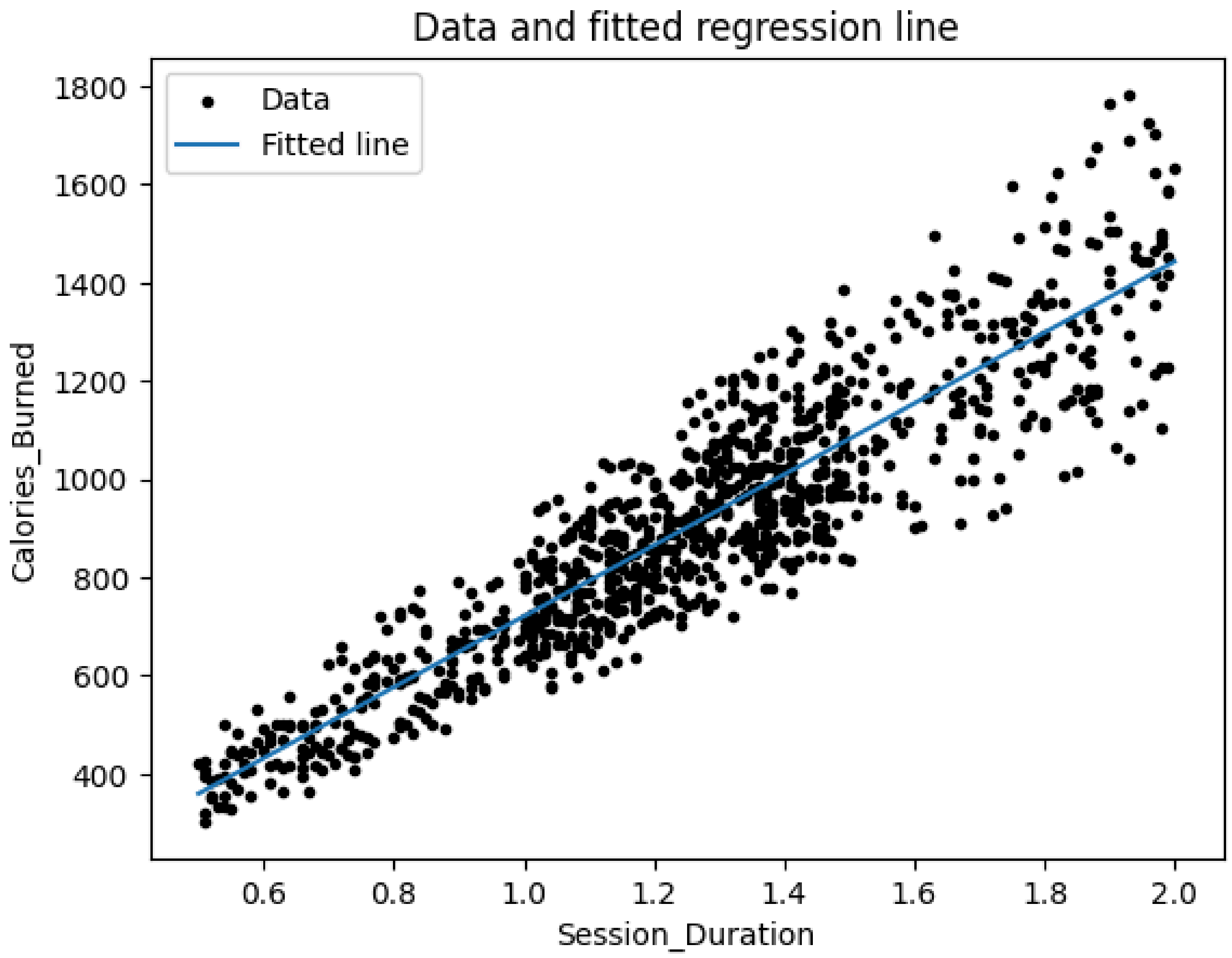
$$y = a + bx + \varepsilon$$

- y : output / outcome
- x : input variables / predictor
- a : intercept
- b : slope

$$y = -1.4465 + 721.7860x$$

Parameter	Estimate	Std Error	Interpretation
Intercept	- 1.4465	25.87	Calories at 0 hrs (not meaningful)
Session Duration	721.7860	19.75	Each +1 hr → +722 kcal on average

REGRESSION LINE – VISUAL FIT



721.79
Slope

-1.45
Intercept

905.4 kcal
Predicted calories

COEFFICIENT INTERPRETATION

Standard Form

$$y = -1.4465 + 721.7860x$$

Intercept = −1.45: Predicted calories when duration = 0 hours.
Not meaningful in practice (no one exercises for 0 hours).

Mean-Centered Form

$$y = 905.4 + 721.79\,(x - 1.3)$$

When duration = average (1.3 hrs), predicted calories = 905.4.
Makes the intercept directly interpretable.

Key Takeaway: Each additional hour of exercise → ~722 more calories burned on average

CONCLUSION

1. A clear positive linear relationship exists between session duration and calories burned.
2. The OLS model explains the trend well with a slope of 721.79 kcal/hour.
3. At the average duration of 1.3 hours, members burn approximately 905.4 kcal.
4. The mean-centered equation improves interpretability of the intercept.
5. The standard form intercept (−1.45) is statistically not meaningful in practice.

THANK YOU FOR YOUR TIME

github.com/nawar27

munawarrf8@gmail.com

DATA ANALYST PRESENTATION